Status	Finished
Started	Thursday, 30 October 2025, 12:46 PM
Completed	Thursday, 30 October 2025, 12:58 PM
Duration	12 mins 45 secs
Marks	2.00/3.00
Grade	**2.00** out of 3.00 (**66.68**%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade**.

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question **1**

Incorrect

Mark 0.00 out of 1.00

Consider the recurrence relation

$$12 \cdot a_n + 2 \cdot a_{n-1} + 4 \cdot a_{n-2} = 0\,.$$

Let $\{a_n\}_n$ and $\{b_n\}_n$ be two numerical sequences. We say that a_n *is a solution* if the sequence $\{a_n\}_n$ satisfies the recurrence relation, and likewise for b_n.
Select every statement below which is true, and leave the false statements unchecked.

- [ ] If a_n and b_n are two solutions, then $d_n = a_n + 3\, b_n$ is also a solution.
- [ ] If a_n is a solution, and a_0=3 and a_1=3, then there exists at least one index n such that a_n is not an integer.
- [ ] If a_0=3, a_1=-5, then a_n can not be a solution.
- [ ] a_n and b_n are solutions, and if a_3=b_3, then a_n=b_n for each n≥1.
- [ ] If a_n is a solution, and a_3=a_4=a_5=0, then a_n=0 for each n≥1.
- [ ] The zero sequence a_n=0 is not a solution.
- [x] None of the above

Question **2**

Correct

Mark 1.00 out of 1.00

Let $(D_3, \cdot)$ be the dihedral group of symmetries of the regular triangle. We label the vertices of the triangle $1, 2, 3$ in a counter-clockwise manner. The symmetry $r1$ is rotation by 120 degrees, sending vertex 1 to vertex 2. The symmetry $r2$ is rotation by 240 degrees. The element $t1$ is the reflection symmetry that fixes vertex 1, and likewise for $t2$ and $t3$.

The multiplication table for this group looks like

$$\begin{bmatrix} \text{id} & \text{r1} & \text{r2} & \text{t1} & \text{t2} & \text{t3} \\ \text{r1} & . & \text{id} & . & . & \text{t1} \\ \text{r2} & . & \text{r1} & . & . & \text{t2} \\ \text{t1} & . & \text{t2} & . & . & \text{r1} \\ \text{t2} & \text{t1} & \text{t3} & \text{r1} & \text{id} & \text{r2} \\ \text{t3} & \text{t2} & \text{t1} & \text{r2} & \text{r1} & \text{id} \end{bmatrix}$$

However, we have left out 9 entries in the places marked by dots.

Find the entire multiplication table for D_3.

Syntax help: Remember that we compose symmetries from left to right. So if a and b are two symmetries, then the composite $a \cdot b$ is the symmetry that first applies a, and then b.

$r2 \cdot r1$ = id

$t1 \cdot r1$ = t3

$r1 \cdot r1$ = r2

$r2 \cdot t2$ = t1

$t1 \cdot t2$ = r2

$r1 \cdot t2$ = t3

$r2 \cdot t1$ = t3

$t1 \cdot t1$ = id

$r1 \cdot t1$ = t2

Your last answer was interpreted as follows:

$$\text{id}$$

Your last answer was interpreted as follows:

$$\text{t3}$$

Your last answer was interpreted as follows:

$$\text{r2}$$

Your last answer was interpreted as follows:

$$\text{t1}$$

Your last answer was interpreted as follows:

$$\text{r2}$$

Your last answer was interpreted as follows:

$$\text{t3}$$

Your last answer was interpreted as follows:

$$t3$$

Your last answer was interpreted as follows:

$$id$$

Your last answer was interpreted as follows:

$$t2$$

Question **3**

Correct

Mark 1.00 out of 1.00

Let $(P_5, \bullet_5)$ be the group of natural numbers coprime to 5 with group operation given by multiplication modulo 5. The multiplication table for this group looks like

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & . & . & . \\ 3 & . & . & . \\ 4 & . & . & . \end{bmatrix}$$

However, we have left out 9 entries in the places marked by a dot "."

Compute the missing entries in the group multiplication table:

$2 \bullet_5 4$ = 3

$3 \bullet_5 4$ = 2

$4 \bullet_5 4$ = 1

$2 \bullet_5 3$ = 1

$3 \bullet_5 3$ = 4

$4 \bullet_5 3$ = 2

$2 \bullet_5 2$ = 4

$3 \bullet_5 2$ = 1

$4 \bullet_5 2$ = 3

Your last answer was interpreted as follows:

$$3$$

Your last answer was interpreted as follows:

$$2$$

Your last answer was interpreted as follows:

$$1$$

Your last answer was interpreted as follows:

$$1$$

Your last answer was interpreted as follows:

$$4$$

Your last answer was interpreted as follows:

$$2$$

Your last answer was interpreted as follows:

$$4$$

Your last answer was interpreted as follows:

$$1$$

Your last answer was interpreted as follows:

3

Question 4

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 226

Signing code: 20323

Your last answer was interpreted as follows:

$$20323$$

◄ Test 2 (topics 4-6: Languages, Machines, Regular expressions, Correct programs, Relations)

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